

The background is a stylized, low-poly illustration of a hospital corridor. It features medical equipment like monitors, IV stands, and gurneys. There are also figures of medical staff in white coats. The color palette is light, with various shades of gray, white, and pale blue/purple.

Weekly Hospital Respiratory Service Demand Report (2020–2024)

Hospitals often face unpredictable fluctuations in respiratory admissions, causing challenges in staffing, bed allocation, and patient care. This project analyzes historical weekly hospital data to forecast demand and provide actionable recommendations.

Project Title & Problem Statement

Forecasting Weekly Hospital Respiratory Service Demand (2020–2024)

Hospitals often face unpredictable fluctuations in respiratory admissions, causing challenges in staffing, bed allocation, and patient care. This project analyzes historical weekly hospital data to forecast demand and provide actionable recommendations.

Staffing Challenges

Unpredictable respiratory admissions make it difficult to schedule adequate medical personnel

Bed Allocation Issues

Fluctuating demand creates inefficiencies in hospital bed management and utilization

Patient Care Impact

Resource constraints affect the quality and timeliness of respiratory care delivery

Project Objectives & Research Questions

Objectives

- Analyze weekly trends in respiratory-related hospital metrics
- Identify key predictors of respiratory admissions
- Develop a predictive model for weekly demand
- Provide operational recommendations for resource allocation

Research Questions

1. What are the seasonal and yearly trends in respiratory admissions?
2. How do hospital metrics like ICU occupancy correlate with respiratory admissions?
3. Can weekly respiratory admissions be accurately forecasted using historical data?
4. Which operational strategies can improve hospital planning?

Data Exploration & Cleaning

Dataset Description

The dataset includes columns for week Ending Date, Actual beds occupied, ICU occupancy, and other metrics spanning from January 2020 through December 2024.

Data Cleaning Steps

- Missing values handled
- Dates converted to datetime format
- Features created: Week of Year, Month, Year, Lag features, Seasonal indicators

Modelling & Predictive Analytics

Modeling Approach

The analysis employed a train/test split strategy using 2020–2023 data for training and 2024 data for testing. The target variable was number_of_inpatient_beds_occupied.

Evaluation Metrics

Metric	Value
MSE	73642640.91
RMSE	8581.53
MAE	1016.68

Results & Insights

Top Prediction Errors

Week Ending	Actual	Predicted	Error
2024-11-16	508,433	505,953	2,480
2024-11-02	314,640	288,480	2,160
2024-10-26	212,040	206,100	5,940
2024-10-19	194,640	204,920	-1,028
2024-09-28	43,416	41,690	1,726

High-Demand Weeks

Week Ending	Predicted
2024-11-16	505,953
2024-11-02	288,480
2024-10-26	206,100
2024-10-19	204,920
2024-09-28	41,690

Key Insights

- Positive errors indicate underprediction of respiratory admissions
- Negative errors indicate overprediction of respiratory admissions
- Seasonal peaks observed during flu season and pandemic periods
- High-demand weeks accurately predicted for proactive planning

Recommendations

1 Increase staffing during high-demand weeks

Proactively schedule additional respiratory care personnel and support staff during predicted peak periods to ensure adequate patient care and reduce staff burnout.

3 Prepare ICU capacity ahead of high-demand weeks

Coordinate with ICU management to ensure ventilators, monitoring equipment, and specialized staff are available before predicted demand spikes occur.

2 Allocate additional beds during peak periods

Reserve and prepare extra respiratory beds and ICU capacity ahead of high-demand weeks to accommodate surge in admissions without compromising care quality.

4 Plan preventive interventions for seasonal spikes

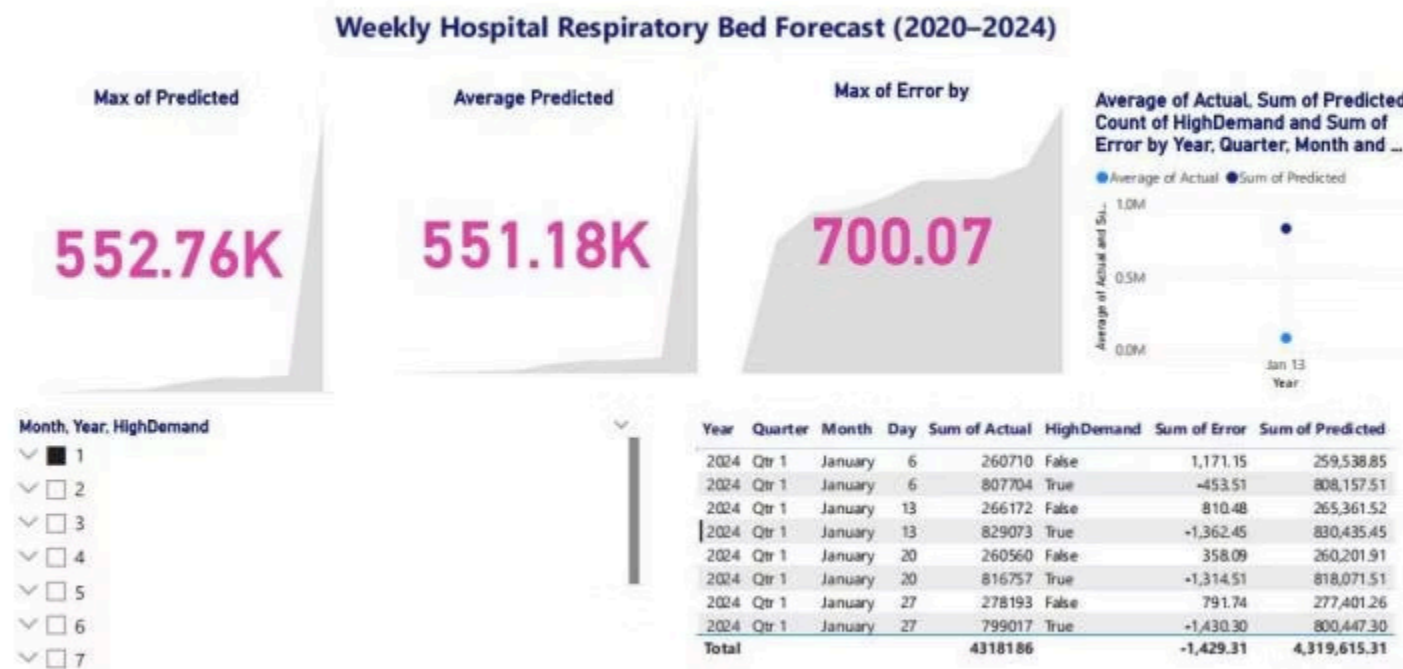
Implement community health initiatives and vaccination campaigns during off-peak seasons to reduce respiratory admissions during predictable seasonal peaks.

Power BI Dashboard

Interactive Dashboard Features

- KPI cards: Average beds, Max predicted beds, High-demand weeks, Top error weeks
- Table: Week-by-week details with actual, predicted, and error values
- Slicers: Year, Month, HighDemand flag for dynamic filtering

The dashboard provides hospital administrators with real-time visibility into respiratory service demand patterns, enabling data-driven decision-making for resource allocation and operational planning.



Dashboard Data Summary

2024 Q1 Performance Metrics

Year	Quarter	Month	Actual	Predicted	Error	High Demand
2024	Q1	January 6	260,710	259,538.85	1,171.15	False
2024	Q1	January 6	807,704	808,157.51	-453.51	True
2024	Q1	January 13	266,172	265,361.52	810.48	False
2024	Q1	January 13	829,073	800,438.45	-1,362.46	True
2024	Q1	January 20	260,560	260,201.91	358.09	False
2024	Q1	January 20	816,757	818,071.51	-1,314.51	True
2024	Q1	January 27	278,193	277,401.26	791.74	False
2024	Q1	January 27	799,017	800,447.30	-1,430.30	True

Total Actual: 4,318,186 | **Total Predicted:** 4,319,615.31 | **Total Error:** -1,429.31

Conclusion

Effective Forecasting

The predictive model successfully forecasts weekly respiratory admissions, enabling hospitals to anticipate demand fluctuations with measurable accuracy.

Proactive Planning

High-demand weeks can be identified ahead of time, allowing hospital administrators to implement operational strategies for resource allocation and staffing.

Actionable Intelligence

Python analysis combined with Power BI dashboard provides hospital administrators with comprehensive, data-driven insights for informed decision-making and improved patient care outcomes.

By leveraging predictive analytics and interactive dashboards, hospitals can transform historical respiratory service data into strategic operational advantages, ensuring adequate staffing, optimal bed allocation, and enhanced patient care delivery throughout the year.